

Kivan Polimis

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Full-stack data scientist (PhD) with 13+ years designing metrics frameworks, production pipelines, and forecasting systems for cross-functional teams across health, policy, and technology. Fluent in Python and R; strong SQL and statistical modeling; production experience with Snowflake, dbt, and cloud data infrastructure (AWS, Azure). Turns high-stakes, ambiguous problems into durable systems: predictive models, shared infrastructure, and cross-functional processes built to outlast any single project. Drives data-informed decisions by making complex, multi-source systems legible for government agencies, clinical programs, consumer analytics and B2B clients.

EMPLOYMENT

2021–Present Senior Data Scientist, Karna, Atlanta, GA 30345

- Engineered a document intelligence microservice that automated clinical document extraction (Azure Document Intelligence, Textract) and AI-powered synthetic data generation (GPT-4); deployed on ECS with schema enforcement and data validation to protect downstream quality.
- Built compliant Snowflake data infrastructure and SQL-based ETL pipelines using Python, R, and shell scripting to ingest multi-source clinical datasets and surface opioid use trends, program utilization metrics, and drug co-occurrence patterns supporting national federal opioid response planning.
- Designed and deployed a HIPAA/NIST-compliant ML pipeline for maternal CVD risk stratification on Azure, with automated drift monitoring and a Flask API; migrated from SQLite to PostgreSQL in production. Applied causal inference (ITS and diff-in-differences) to measure the program's impact on emergency care utilization.
- Partnered with data science, research and evaluation, and health communications teams to design and run A/B message testing for CDC NCIRD's COVID Vaccine Task Force; tested message variants across hesitancy segments, political ideology, and demographics using ordered logistic regression; findings directly shaped national vaccine communication decisions.

Awards: President's Award (2023), Technical Achievement Award (2023), Community Engagement Award (2022), Outstanding Publication runner-up (2022), Technical Achievement runner-up (2024)

2021–Present Lecturer (part-time), University of Washington, Department of Sociology, Seattle, WA 98195

- Designed and taught applied labs on behavioral and digital trace data analysis (Google Trends, Fitbit, sensor data), data visualization, NLP, and algorithmic fairness; challenged students to interrogate models for potential bias from training data, historical inequities, and deployment context.
- Designed and ran in-class randomized experiments (A/B tests) covering random assignment, hypothesis testing, and power analysis, demonstrating how controlled experimentation underpins data-driven product and policy decisions and where purely observational analysis falls short.

2020–2021 Senior Policy Advisor, Office of Harris County Judge Lina Hidalgo, Houston, TX 77002

- Automated COVID-19 surveillance reporting by replacing hand-transcribed web data and manual multi-source joins with a reproducible R pipeline ingesting DSHS, HCPH, and national datasets, reducing processing latency by 90% and enabling real-time data quality for policymakers.
- Applied statistical modeling to evaluate county-wide housing initiatives, projecting \$6 million in annual savings from permanent supportive housing proposals.

2020–Present Principal and Founder, Atlas Analytics, Houston, TX 77003

- Built a Python/SQL entity resolution pipeline joining fragmented SBA datasets with merchant registries via a tiered probabilistic matching algorithm (86% recall on a manually labeled hold-out set), enabling downstream customer analytics and market segmentation.
- Built data pipelines integrating a 2.2M-member consumer transaction panel (Earnest Analytics), device mobility data (SafeGraph), and COVID policy records; analyzed user acquisition and retention cohorts to quantify COVID-19's long-term impact on consumer behavior across 12 categories including restaurant delivery; acknowledged for research assistance in Oblander & McCarthy, Marketing Science (2023).

2018–2020 Data Scientist, MAANA, Houston, TX 77002

- Partnered with solutions architects to engineer and deploy containerized ML microservices (Docker, Kubernetes) powering a real-time vessel routing optimizer across global multi-port routes, with ingestion

pipelines parsing Q88 XML and integrating live weather, war-risk, and port data to cut fleet planning from hours to minutes; cost forecasting on historical trades projected \$100M in annual savings.

2017–2018 Postdoctoral Researcher, Università Bocconi, Milan, Italy 20136

Carlo F. Dondeña Centre for Research on Social Dynamics and Public Policy

- Engineered multi-source data fusion pipelines from social media APIs and digital trace data to produce high-frequency population estimates and evaluate policy outcomes at scale.

DATA SCIENCE & ENGINEERING PROJECTS

Multi-Sport Data & Analytics Platform

- Architected a production-grade sports prediction web app and FastAPI service serving calibrated win probabilities for NBA, NCAA, and NFL games; designed ETL pipelines ingesting from The Odds API, KenPom, and Sports-Reference into a Postgres backend with schema enforcement and lineage tracking.
- Built an NBA data platform (nba_api → Airflow → dbt → Postgres) implementing a medallion architecture with dbt schema tests, source freshness checks, and Airflow-automated data quality validation.

Job Intelligence Platform

- Architected a self-hosted, containerized job intelligence platform (Docker Compose + Ollama) that aggregates postings from multiple job boards daily, enriches each listing with a local LLM, and ranks by résumé fit using sentence embeddings and cosine similarity, delivering plain-language match narratives and a transparent Apply Priority heuristic on fully local infrastructure.
- Developed a capability-matrix scoring layer that maps job requirements to structured skill ratings and generates per-match talking points; a self-service data product that converts a judgment-heavy screening process into a quantified, reproducible workflow with end-to-end data quality validation.

TECHNICAL SKILLS

Instructor	Software Carpentry Instructor, Data Carpentry Instructor
Cloud/DevOps	AWS (ECR, SageMaker, Lambda, S3), Azure (ACR, AKS, Azure Functions), Docker, Git, CI/CD
Data Engineering	ETL/ELT pipeline design, distributed data systems (Snowflake, dbt, Airflow), PostgreSQL, SQL Server, SQLite, schema enforcement, data lineage, medallion architecture
Programming	Python, SQL, R, shell scripting
Project Tools	Jira, GitHub Projects

EDUCATION

- 2017 Ph.D.**, Computational Social Science, University of Washington
National Science Foundation Graduate Research Fellowship recipient
Center for Statistics and the Social Sciences: Certification in Social Statistics
Center for the Study of Demography and Ecology Pre-doctoral Fellow
Data Science for Social Good Fellow
- 2012 M.A.**, Sociology, University of North Carolina at Chapel Hill
Carolina Population Center Pre-doctoral Fellow
- 2009 B.A.**, Sociology, Princeton University

SELECT PUBLICATIONS

- 2020** Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. “Combining social media and survey data to nowcast migrant stocks in the United States”. *Population Research and Policy Review*.
- 2020** Salganik, Matthew, Ian Lundberg, Alexander T. Kindel, Caitlin E. Ahearn, et al.. “Measuring the predictability of life outcomes with a scientific mass collaboration”. *Proceedings of the National Academy of Sciences*.
- 2019** Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. “The impact of Hurricane Maria on out-migration from Puerto Rico: Evidence from Facebook”. *Population and Development Review*.